Route G-R12: Local Correlated Preparation and the Energy Cost of Its Mediator

An added controlled interaction; unchanged background contact

9 October 2026

Abstract

The G-R9b density-contact obstruction does not follow from translation symmetry alone. A local occupation-dependent hopping realizes the desired dark-clock preparation block without inserting a momentum projector. Keeping the old contact, one fixed pulse gives unconditional fidelities above 0.997 in the inherited fixtures. A finite, 21-state mediator model then exposes a leading Stark term and high-energy switching residue. This is a controlled preparation construction, not autonomous event or time generation; its drive phase, pulse and compensation are explicit.

1 A local interaction outside the density-only family

Use two distinguishable three-node rings, with $S|j\rangle = |j+1 \bmod 3\rangle$, and retain the full G-R10 system Hamiltonian

$$L_\theta = 2I - e^{\mathrm{i}\theta/3}S - e^{-\mathrm{i}\theta/3}S^\dagger, \tag{1}$$

$$H_\gamma = H_0 + \gamma P_0 \otimes P_0, \quad H_0 = L_{\pi/2} \otimes I + I \otimes L_{\pi/2}. \tag{2}$$

All nine product states remain present. In the site basis define

$$D = \sum_{j=0}^{2} P_j \otimes P_j, \quad K = L_0 \otimes I + I \otimes L_0, \qquad C = \frac{\mathrm{i}}{\sqrt{2}}[K, D]. \tag{3}$$

D detects coincidence, not an external density or Chen's E_d. For configurations $x = (j, l)$ and $y = (j', l')$, put $d_x = \delta_{jl}$. Then

$$C_{xy} = \frac{\mathrm{i}}{\sqrt{2}} K_{xy}(d_y - d_x). \tag{4}$$

Thus C contains only single-ring nearest-neighbour hops which change coincidence. There are 12 undirected edges (24 directed entries), not an arbitrary projector onto selected Fourier modes. Graph locality is not a derivation of physical space. $C = C^\dagger$ because $[K, D]$ is anti-Hermitian. Both K and D commute with joint translation $\mathsf{T} = S \otimes S$ and the swap X, so C does too.

K is a *new zero-flux hopping quadrature*, not the existing H_0. It is not generated merely by retiming the original contact. We add gC during preparation and account for its switching separately.

2 Exact preparation block and the precise symmetry correction

Let $F_{jk} = e^{2\pi \mathrm{i} jk/3}/\sqrt{3}$ and use the fixed Fourier basis. The joint-translation sector $a + b = 2 \bmod 3$ is exactly $\mathcal{T} = \mathrm{span}(|02\rangle, |11\rangle, |20\rangle)$. Writing Π for its projector, direct Fourier transformation gives

$$K_{\mathcal{T}} = \mathrm{diag}(3, 6, 3), \quad D_{\mathcal{T}} = \frac{1}{3}\begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}, \quad C_{\mathcal{T}} = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 & -\mathrm{i} & 0 \\ \mathrm{i} & 0 & \mathrm{i} \\ 0 & -\mathrm{i} & 0 \end{pmatrix}. \tag{5}$$

No coupling leaves this sector under C. Set

$$|M\rangle = |11\rangle, \quad |A\rangle = \frac{|02\rangle + |20\rangle}{\sqrt{2}}, \quad |D_+\rangle = \frac{1}{\sqrt{3}}|A\rangle - \sqrt{\frac{2}{3}}|M\rangle. \tag{6}$$

Then $C_{\mathcal{T}} = \mathrm{i}(|M\rangle\langle A| - |A\rangle\langle M|)$ and $H_{0,\mathcal{T}} = 4I$. Hence, for $\gamma = 0$ and up to an overall phase,

$$e^{-\mathrm{i}s(H_0 + gC)}|M\rangle = \cos(gs)|M\rangle - \sin(gs)|A\rangle, \quad \beta = \arcsin(1/\sqrt{3}), \quad gs_* = \beta, \tag{7}$$

which equals $-|D_+\rangle$ at s_*. For any area $a = gs$,

$$F_0(a) = \cos^2(a - \beta), \qquad F_0(-\beta) = 1/9. \tag{8}$$

The coupling sign is therefore physical with the original state and readout held fixed. At $a = \beta + \pi/2$ the target fidelity vanishes: continued control is not storage.

What G-R9b does and does not rule out. Its $h(j-l)$ density contacts give circulant blocks and conserve their common bright component. Equation (5) is not circulant, even though it commutes with T and X. Joint translation acts as a scalar throughout $\mathcal{T}$, so it cannot force a particular matrix within that sector. The 2/3 obstruction therefore applies to the specified contact family, not to every translation-invariant interaction. The old rank-one 4/9 bound remains valid and unchanged. Also $\Pi[H_0, D]\Pi = 0$: replacing the new K by the old resonant kinetic term would remove precisely the desired block.

An exact occupation-dependent Peierls identity further characterizes the new operator. For $K_\chi = e^{\mathrm{i}\chi D} K e^{-\mathrm{i}\chi D}$,

$$C = -\frac{K_\chi - K_{-\chi}}{2\sqrt{2}\sin\chi}, \qquad \sin\chi \neq 0. \tag{9}$$

It follows entry by entry since $d_x - d_y \in \{0, \pm 1\}$. Adding this difference while retaining H_0 is a new interaction. Conjugating every state, operator and probe together would instead be only a basis change.

3 Keep the original contact and count the controlled pulse

The actual direct-pulse test uses $H_{\mathrm{on}} = H_\gamma + gC$, not the isolated two-state block. Inherited parameters are $g = 0.1$ and $\alpha = 0.25$ per assumed update. Before evaluating any γ, fix

$$n = \mathrm{round}\,\frac{\beta}{g\alpha} = 25, \qquad s_f = n\alpha = 6.25 \quad (s_* = 6.15479709\ldots). \tag{10}$$

There is no pulse optimization for individual fixtures. s remains the declared evolution parameter, not physical seconds. Exact 9×9 exponentials give the following *unconditional* results:

γ	F_D	probability outside $\mathcal{T}$	net work on system
0	0.999909367	0	0
0.09	0.997140947	$1.78685\,10^{-4}$	−0.00992608
0.03	0.999600916	$1.97636\,10^{-5}$	−0.00333098
0.01	0.999875084	$2.19518\,10^{-6}$	−0.00111089

The original vertex contact breaks the sector closure, so leakage is retained rather than projected away. During the hold, the expectation of H_{on} is constant. Sudden switching gives the exact ledger

$$W_{\mathrm{on}} = g\langle M|C|M\rangle = 0, \quad W_{\mathrm{off}} = -g\langle\psi_f|C|\psi_f\rangle, \tag{11}$$

$$\langle H_\gamma\rangle_f - \langle H_\gamma\rangle_i = W_{\mathrm{on}} + W_{\mathrm{off}}, \qquad \langle H_\gamma\rangle_i = 4 + \gamma/9. \tag{12}$$

Negative work means energy is delivered to the controller. It does not mean that the pulse apparatus, switching precision or resetting is free. The Hamiltonian before and after the pulse is the same H_γ.

Why this does not contradict the old energy bound. For the inherited positive γ fixtures, the $E = 4$ eigenspace is $P_D = |D_+\rangle\langle D_+| + |D_-\rangle\langle D_-|$. Any system-only unitary commuting with H_γ preserves its weight, so

$$[U, H_\gamma] = 0 \quad \Longrightarrow \quad |\langle D_+|U|M\rangle|^2 \le \langle M|P_D|M\rangle = 2/3. \tag{13}$$

Our pulse is not such a unitary; the full work ledger is essential. An autonomous energy receiver would have to match the full energy distribution, not merely its mean.

Connection to the existing record. Swap symmetry leaves the antisymmetric D_- amplitude zero. Thus the unchanged G-R11 reader now succeeds with probability exactly F_D, not 2/3. Conditional on success its unread output is still $(57/82)|D_+\rangle\langle D_+| + (25/82)|D_-\rangle\langle D_-|$. The measurement disturbance has not disappeared. After pulse removal, F_D stays constant because D_+ is an eigenstate of H_γ; this does not make the entire imperfect output state stationary.

4 A finite local mediator, including its leading Stark term

To check more than a prescribed effective operator, introduce one auxiliary configuration state for each of the 12 undirected edges in Eq. (4). These are explicit additional degrees of freedom, initially empty, with common energy $4+\Delta$. For edge $e=(x,y)$, $x<y$, choose

$$B_{ex}=\Omega, \quad B_{ey}=-\frac{g\Delta C_{xy}}{\Omega}, \quad \Omega=\sqrt{\frac{g\Delta}{\sqrt{2}}}, \qquad B_{ez}=0 \ (z\neq x,y). \tag{14}$$

Both nonzero amplitudes have modulus Ω. Their relative phase is supplied; the construction does not derive its sign from symmetry. The full 21×21 hold Hamiltonian is

$$\mathcal{H}_{\mathrm{on}}=\begin{pmatrix} H_\gamma+V_c & B^\dagger \\ B & (4+\Delta)I_{12}\end{pmatrix}, \quad \mathcal{H}_{\mathrm{off}}=H_\gamma\oplus(4+\Delta)I_{12}. \tag{15}$$

Every auxiliary exit is retained in the exact propagation.

Eliminating auxiliaries from the eigenvalue equation gives the exact energy-dependent Schur operator

$$H_{\mathrm{low}}(E)=H_\gamma+V_c-\frac{B^\dagger B}{\Delta-(E-4)}, \tag{16}$$

$$H_{\mathrm{low}}(E)=H_\gamma+V_c-\frac{B^\dagger B}{\Delta}+R(E), \quad \|R(E)\|\leq\frac{\|B^\dagger B\||E-4|}{\Delta(\Delta-|E-4|)} \quad (|E-4|<\Delta). \tag{17}$$

This controlled low-energy expansion alone does not certify a suddenly switched initial state. The diagonal terms already matter at leading order. A coincident configuration has four mediator edges, a noncoincident one two. Consequently,

$$-\frac{B^\dagger B}{\Delta}=gC-\sqrt{2}g(I+D). \tag{18}$$

The off-diagonal equality follows from $-B_{ex}^*B_{ey}/\Delta=gC_{xy}$; the degree count gives the diagonal. Thus $-\sqrt{2}gD$ is not a higher-order nuisance that can be omitted. An explicitly additional compensation

$$V_c=+\sqrt{2}gD \tag{19}$$

cancels it to this order, leaving the irrelevant low-sector identity shift $-\sqrt{2}gI$. The compensation coefficient is derived from the balanced mediator couplings, not fitted to a desired final fidelity.

Density-assisted hopping with controllable Peierls phases has been realized using driven cold atoms [1]. This supports the physical operator class, not our particular 12 auxiliary states, compensation, numerical scales, or autonomous implementation. The construction here is a finite mediator candidate with local edge support, not a completed laboratory device or unique first-principles action.

5 High-energy residue changes the physical cost

Fix $\gamma = 0.03$, the same $g = 0.1$ and the same 25-update pulse. The three declared detunings test the mediator, not a fit of masses or pulse durations. Without compensation, the effective target fidelity is 0.95183145; with compensation it is 0.99960092. The full model gives

V_c present	Δ	full F_D	P_{aux}	E_{aux}	W_{net}
no	20	0.93561813	0.0249881	0.599715	0.496818
no	40	0.94438231	0.0125617	0.552714	0.499559
no	80	0.94935599	0.0042897	0.360335	0.340302
yes	20	0.98003504	0.0188301	0.451921	0.373277
yes	40	0.99051139	0.00886303	0.389973	0.351192
yes	80	0.99631112	0.00333585	0.280212	0.263537

Here $F_D = |(\langle D_+|, 0_{12})\Psi_f|^2$, not a fidelity renormalized after rejecting auxiliaries, and

$$E_{\mathrm{aux}} = (4 + \Delta)P_{\mathrm{aux}}, \qquad P_{\mathrm{sys}} + P_{\mathrm{aux}} = 1. \tag{20}$$

With $\mathcal{V} = \mathcal{H}_{\mathrm{on}} - \mathcal{H}_{\mathrm{off}}$, the complete switching ledger is

$$W_{\mathrm{on}} = \langle \Psi_i|\mathcal{V}|\Psi_i\rangle = \begin{cases} \sqrt{2}g/3 & \text{with compensation,} \\ 0 & \text{without,} \end{cases} \tag{21}$$

$$W_{\mathrm{off}} = -\langle \Psi_f|\mathcal{V}|\Psi_f\rangle, \tag{22}$$

$$W_{\mathrm{net}} = W_{\mathrm{on}} + W_{\mathrm{off}} = \langle \mathcal{H}_{\mathrm{off}}\rangle_f - \langle \mathcal{H}_{\mathrm{off}}\rangle_i. \tag{23}$$

The expectation of $\mathcal{H}_{\mathrm{on}}$ is conserved throughout the hold. Thus there is no missing energy: the controller supplies the positive net work displayed here. In the compensated $\Delta = 80$ case, only about 0.334% remains auxiliary, yet it carries energy 0.2802 (excess above the common energy 4 is $\Delta P_{\mathrm{aux}} = 0.2669$). The direct effective-pulse work -0.003331 cannot replace this ledger.

The origin is the sudden boundary. In the frame shifted by energy 4, writing the system and auxiliary amplitudes as $a(s), b(s)$ gives exactly

$$b(s) = -\mathrm{i}\int_0^s e^{-\mathrm{i}\Delta(s-u)} B a(u)\, du. \tag{24}$$

The dressed low-energy component and a rapidly oscillating boundary component are both present. Since $\|B\|$ scales as $\sqrt{g\Delta}$, an auxiliary probability of order g/Δ can carry energy of order g. Small population alone supplies no small-energy bound. This is not a claim of a monotonic or universal asymptotic coefficient.

Extending the existing readout by assigning every auxiliary branch to failure retains total probability one. Its success probability remains the unconditional F_D in the table. No bath, unreported sink or postselected energy accounting has been introduced.

6 Physical interpretation and the next decision

Three tasks must remain separate. Transport to a single outlet, coherent clock preparation and durable event recording have different criteria. G-R9b's chiral transfer reaches $|02\rangle$, a product state whose remote conditional density matrix $\rho_{B|\phi} = |2\rangle\langle 2|$ does not vary with A's phase reading. Its full cycle returns to the input after $3t_*$; direction is not irreversibility. The prepared D_+ instead supports G-R10's nontrivial conditional orbit. Neither construction alone generates an autonomous saved event.

The other scope correction concerns carriers: propagation does not require a gapless band. For example, $E(k) = \Delta_c + 2J(1 - \cos k)$ has $v_g = 2J \sin k/\hbar \neq 0$ despite $\Delta_c > 0$ (unit lattice spacing). This example does not derive a spatial network or a passing trigger; it only rejects a needless gaplessness prerequisite.

What is established. Equations (3)–(5) realize the preparation generator by local conditional hopping rather than a hand-inserted mode projector. The original contact is retained in all actual nine-state fixtures. Equations (15)–(23) give an explicit finite relay, its Stark compensation and an all-channel energy ledger. The high fidelity is a model prediction, not measured hardware performance. The inherited ring flux, new hopping quadrature and phase, balanced relay amplitudes, auxiliary detuning, pulse and controller remain physical inputs. Locality and symmetry permit them but do not select them uniquely. This is not an unconditional first-principles derivation.

A bounded next choice. The local-operator existence and finite-mediator audit can close here. For a controlled realization, choose one physical mediator/control implementation and test a single smooth on/off protocol, counting auxiliary energy and the compensation work together. For autonomous events, replace the imposed pulse by a specified passing trigger and include reflection, return and its energy. These are alternatives, not new mandatory papers before using the bounded result. No new mass fit, universal E_d law, time arrow or Route-F promotion follows.

Reproducibility. The new verifier has 189 passing checks: local support, exact block, symmetries, finite Peierls identity, original-contact dynamics, full work ledgers, Stark terms, and all auxiliary/readout branches. Matrix exponentiation and spectral propagation agree independently. The G-R9b and G-R11 code, cards and result hashes are pinned unchanged; their valid restricted statements are preserved, not rewritten as new claims. The source files are under Route G:

```
data/CORRELATED_PREPARATION_CARD.json
code/verify_gr12_correlated_preparation.py
output/gr12_correlated_preparation.json
```

References

[1] F. Görg et al., *Realization of density-dependent Peierls phases to engineer quantized gauge fields coupled to ultracold matter*, Nature Physics **15**, 1161–1167 (2019), arXiv:1812.05895, doi:10.1038/s41567-019-0615-4.

7 G-R13 addendum: one fixed smooth switching pulse

11 October 2026. Sections 1–6 retain the G-R12 checkpoint; this addendum completes its single-pulse test, not a pulse search.

Fix the same 21-state model, $\gamma = 0.03$, $\Delta = 80$ and $g = 0.1$. Write $B_0 = B$ from Eq. (15) and define

$$V_B = \begin{pmatrix} 0 & B_0^\dagger \\ B_0 & 0 \end{pmatrix}, \qquad V_{c,0} = (\sqrt{2}gD) \oplus 0_{12}. \tag{25}$$

Choose just one envelope, without changing the peak or optimizing its area:

$$f(s) = \begin{cases} \sin^2(\pi s/T), & 0 < s < T, \\ 0, & \text{otherwise}, \end{cases} \qquad \mathcal{H}(s) = \mathcal{H}_{\text{off}} + f(s)V_B + f(s)^2 V_{c,0}, \tag{26}$$

$$\int_0^T gf(s)^2\, ds = \frac{3gT}{8} = g\tau = 0.625, \qquad \tau = 6.25, \qquad T = \frac{8}{3}\tau = \frac{50}{3}. \tag{27}$$

The initial state is $(F \otimes F)|11\rangle \oplus 0_{12}$ as before. The envelope is compact C^1, not C^∞: $f = f' = 0$ at both ends. The continuously prescribed s is an input, not derived physical time; T/α is not an integer and is not rounded to an update count. Equal area means equal *leading effective preparation area*: $\int f\, ds = T/2$ is not the old relay-envelope area τ. The duration therefore increases by $8/3$; equal duration is not claimed.

Compensation must follow the square of the relay envelope. Equation (18) gives the instantaneous leading low-energy identity

$$-\frac{f^2 B_0^\dagger B_0}{\Delta} + f^2\sqrt{2}gD = gf^2 C - \sqrt{2}gf^2 I. \tag{28}$$

A compensation proportional to f would not cancel the Stark term. Equation (28) explains the chosen area but does not replace time-ordered full-model propagation. In particular, finite-gap and time-dependent corrections are not discarded from the calculation. The low-sector identity shift cannot be removed from the physical work ledger. Only a *constant common* $4I_{21}$ is subtracted for numerical propagation; all reported work uses the full Hamiltonian.

Why the terminal residue can decrease. In that common rotating frame let a, b be the nine low and twelve auxiliary amplitudes. With $A(u) = f(u)B_0 a(u)$ and $b(0) = 0$,

$$b(T) = -\mathrm{i}\int_0^T e^{-\mathrm{i}\Delta(T-u)} A(u)\, du = \frac{\mathrm{i}}{\Delta^2}\int_0^T e^{-\mathrm{i}\Delta(T-u)} A''(u)\, du, \tag{29}$$

$$\|b(T)\| \le \frac{1}{\Delta^2}\int_0^T \|A''(u)\|\, du. \tag{30}$$

Two integrations by parts have no boundary terms because $A = A' = 0$ at the endpoints. The abrupt pulse does not have this cancellation. This is an exact identity for the fixed full solution, not a uniform large-Δ theorem: a uniform derivative bound would also be needed for such a claim. The mediator can be occupied during the pulse and return almost entirely before switch-off; it has not been projected out.

8 Full compensation and switching-work ledger

For the normalized solution $\mathrm{i}\dot{\Psi} = \mathcal{H}(s)\Psi$, the commutator contribution cancels in the energy derivative:

$$\frac{d}{ds}\langle\mathcal{H}\rangle = \langle\dot{\mathcal{H}}\rangle = P_B + P_c, \tag{31}$$

$$P_B = \dot{f}\langle V_B\rangle, \qquad P_c = 2f\dot{f}\langle V_{c,0}\rangle, \qquad W_j = \int_0^T P_j\, ds, \tag{32}$$

$$W_{\mathrm{net}} = W_B + W_c = \langle\mathcal{H}_{\mathrm{off}}\rangle_f - \langle\mathcal{H}_{\mathrm{off}}\rangle_i. \tag{33}$$

There are no additional endpoint quenches and no heat term. Both control channels are counted, including energy delivered back to them. To distinguish signed work from supplied energy, with $[x]_+ = \max(x, 0)$, define separate-port and common-bus quantities

$$W_+^{\mathrm{sep}} = \int_0^T ([P_B]_+ + [P_c]_+)\, ds, \qquad W_-^{\mathrm{sep}} = \int_0^T ([-P_B]_+ + [-P_c]_+)\, ds, \tag{34}$$

$$W_+^{\mathrm{bus}} = \int_0^T [P_B + P_c]_+\, ds, \qquad W_-^{\mathrm{bus}} = \int_0^T [-P_B - P_c]_+\, ds. \tag{35}$$

Either convention obeys $W_+ - W_- = W_{\mathrm{net}}$. The common-bus convention allows instantaneous transfer between controllers; the separate-port positive input does not assume that saving. The square reference is computed per port at each of its two quenches, with the state unchanged across the quench. There is no work during its constant hold.

Quantity (model units)	Compensated square	One smooth pulse
Unconditional F_D	0.996311117	0.998120722
Endpoint P_{aux}	$3.335852\,10^{-3}$	$5.02\,10^{-13}$
Endpoint $(4+\Delta)P_{\mathrm{aux}}$	0.280211579	$4.22\,10^{-11}$
W_{net}	+0.263536868	−0.003314374
W_+^{sep}	0.263585530	0.171251419
W_-^{sep}	0.000048662	0.174565792
$W_+^{\mathrm{sep}} + W_-^{\mathrm{sep}}$	0.263634192	0.345817211
Duration	6.25	16.666666667

For the smooth pulse, the explicit components are

$$W_B = -0.0381709643, \qquad W_c = +0.0348565906. \tag{36}$$

Thus compensation is not free. The common-bus positive/returned work is 0.132890641/0.136205015. The sampled maximum auxiliary probability during the pulse is 0.00184626, not its terminal value. Energy above the common level 4 at the endpoint is $\Delta P_{\mathrm{aux}} = 4.02\,10^{-11}$; the net work is independent of that energy-origin convention.

All 21 amplitudes contribute to normalization and energy. The fixed G-R11 reader assigns auxiliaries to failure; success remains the unconditional F_D. Its unread-success state still has the exact 57/82 versus 25/82 dark-state mixture. After control removal the fidelity and auxiliary probability stay constant, not the entire state.

9 What this test closes, and what it does not

Answer for this fixed model. Yes: one prescribed smooth pulse simultaneously improves preparation fidelity, lowers terminal high-energy residue, and lowers both signed net work and separate-port positive input. This is not solely an apparent gain from subtracting recovered work. However, the longer duration and larger absolute energy throughput prevent a claim that every resource cost has improved. Negative net work draws energy from the supplied input state; its preparation, controller, memory reset and any dissipation are not free. There is no bath or hidden energy sink.

For a transparent *conditional* hardware diagnostic, assume the same recovery fraction $0 \le \eta \le 1$ and constant overhead power p_0 for both protocols. Their cost in these assumptions is

$$\mathcal{C} = W_+^{\mathrm{sep}} - \eta W_-^{\mathrm{sep}} + p_0 T. \tag{37}$$

The smooth pulse wins only when

$$\begin{aligned} p_0 &< \frac{W_{+,\square}^{\mathrm{sep}} - W_{+,\mathrm{sm}}^{\mathrm{sep}} + \eta(W_{-,\mathrm{sm}}^{\mathrm{sep}} - W_{-,\square}^{\mathrm{sep}})}{T - \tau} \\ &= 0.008864075 + 0.016753645\,\eta. \end{aligned} \tag{38}$$

No η or p_0 has been measured or fitted. Extra waveform-dependent losses could invalidate this simple cost model. This inequality is a decision criterion in model energy per model time, not a hardware efficiency prediction or a new scan.

Numerical checks are refinements of the same physical pulse. The augmented full 21-component Schrödinger equation and two work integrals are integrated with DOP853 at tolerances $(2\,10^{-12}, 2\,10^{-14})$ and $(2\,10^{-13}, 2\,10^{-15})$. An independent unitary two-exponential propagator uses

$$c_{1,2} = \frac{1}{2} \mp \frac{\sqrt{3}}{6}, \quad a_{1,2} = \frac{3 \mp 2\sqrt{3}}{12}, \quad H_j = \mathcal{H}(s + c_j h) - 4I, \tag{39}$$

$$U_h = e^{-\mathrm{i}h(a_1 H_1 + a_2 H_2)} e^{-\mathrm{i}h(a_2 H_1 + a_1 H_2)}. \tag{40}$$

At $1024, 2048, 4096$ steps its final-state errors relative to the refined ODE solution are approximately $6.35\,10^{-9}, 3.85\,10^{-10}, 2.39\,10^{-11}$, consistent with fourth order. The two finest terminal auxiliary probabilities agree within 0.04%; only a few significant figures of the tiny residue are reported. Independent 10001/20001-point power quadratures agree with integrated work; the first law is checked along the entire path, not just its ends. The coarse ODE tolerance was tightened after an initial state-difference check missed 10^{-10}; no physical parameter or pulse was changed.

All 70 G-R13 checks pass. G-R12 code, card and numerical-result hashes remain pinned to checkpoint `7d6d425`; its historical derivation is retained in Sections 1–6 of this same document. The new artifacts are

```
data/SMOOTH_SWITCHING_CARD.json
code/verify_gr13_smooth_switching.py
output/gr13_smooth_switching.json
```

The bounded single-pulse question is closed. A next physical choice is to calibrate an actual controller against Eq. (38), or, if autonomy is the goal, replace the imposed pulse by a specified passing trigger with its energy and return channels. Neither is a new mandatory mathematical prerequisite. No pulse optimization, emergent seconds, universal E_d law, mass fit or Route-F gate is promoted.